

GMX · SECURITY REVIEW

GMX

OpenInterest Updates.

A security review of OpenInterest Updates.

CLIENT	PROJECT	AUDITORS	AUDIT WINDOW	FINAL REPORT
GMX	OpenInterest Updates	Guardian	December 20 – December 21, 2025	December 29, 2025

§ 00

Contents.

15 pages — overview, confidence ranking, scope, 5 findings, disclaimer, and about Guardian.

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§ 01

Overview.

Guardian reviewed OpenInterest Updates across a single review round. The final report includes 2 Medium, 3 Low findings.

AUDIT FIRM	METHODOLOGY
Guardian	Smart contract · Manual · Main Review
CLIENT	
GMX	
PROJECT	
OpenInterest Updates	
PREPARED BY	
Guardian	
AUDIT WINDOW	
December 20 – December 21, 2025	
FINAL REPORT	
December 29, 2025	

1.1 Confidence Ranking

5 / 5

Very High Confidence

Guardian assigns a Confidence Ranking of 5 out of 5, combining finding severity, remediation outcome, and the reviewer’s assessment of latent risk. The full scale is on the next page.

1	2	3	4	5
VERY LOW CONFIDENCE	LOW CONFIDENCE	MODERATE CONFIDENCE	HIGH CONFIDENCE	VERY HIGH CONFIDENCE

1.2 Findings at a Glance

SEVERITY	TOTAL	PENDING	ACK	PARTIAL	RESOLVED
■ Critical	0	0	0	0	0
■ High	0	0	0	0	0
■ Medium	2	0	2	0	0
■ Low	3	0	3	0	0
■ Informational	0	0	0	0	0
All severities	5	0	5	0	0

§ 02

Confidence Ranking.

Guardian assigns a project-level confidence score using the scale below. The ranking combines finding severity, remediation outcome, and the reviewer's assessment of latent risk.

CONFIDENCE RANKING	DEFINITION AND RECOMMENDATION	RISK PROFILE
5 Very High Confidence ASSIGNED	Codebase is mature, clean, and secure. No High or Critical vulnerabilities were found. Follows modern best practices with high test coverage and thoughtful design. Recommendation: code is highly secure at time of audit.	0 High/Critical findings and few Low/Medium severity findings.
4 High Confidence	Code is clean, well-structured, and adheres to best practices. Design patterns are sound and test coverage is strong. Recommendation: suitable for deployment after remediations; consider periodic review with changes.	0-1 High/Critical findings per engagement week and little to no Medium severity findings.
3 Moderate Confidence	Code is functional but shows gaps in defensive programming or test coverage. Recommendation: remediate the findings and re-review the changes before deployment.	Some High/Critical findings, with various Medium severity findings.
2 Low Confidence	Recurring issues suggest systemic gaps rather than isolated mistakes. Recommendation: remediate, then commission a follow-up review at a frozen commit.	Over 6 High and Critical findings across the engagement.
1 Very Low Confidence	The volume and severity of findings indicate the codebase is not ready for production. Recommendation: substantial rework, then a full re-review.	Over 12 High and Critical findings across the engagement.

§ 03

Scope & Classifications.

Guardian reviewed OpenInterest Updates.

3.1 Severity Matrix

	IMP · HIGH	IMP · MED	IMP · LOW
LIK · HIGH	Critical	High	Medium
LIK · MED	High	Medium	Low
LIK · LOW	Medium	Low	Low

IMPACT · HIGH

Significant loss of assets, harm to a group of users, or core protocol functionality is disrupted.

IMPACT · MEDIUM

Small amount of funds can be lost, or ancillary functionality is affected. Users may experience reduced or delayed receipt.

IMPACT · LOW

Unexpected behavior in some functionality that is notable but does not meet the criteria for higher severity.

LIKELIHOOD · HIGH

Possible with reasonable assumptions that mimic on-chain conditions; cost is low compared to amount gained.

LIKELIHOOD · MEDIUM

Possible only in uncommon cases or requires a large amount of capital relative to disruption caused.

LIKELIHOOD · LOW

Unlikely to ever occur in production.

Files in Scope.

FILE	NSLOC	TOTAL
No files were recorded in scope.		

§ 04

Findings.

Grouped by round. Visible IDs reset by severity within each round, so a finding is named the same way in the report as it was in the channel it was written in.

ID	TITLE	CATEGORY	SEVERITY	STATUS	PAGE
MAIN REVIEW					
M-01	Funding Charged Is Not Accurate To OI In Tokens	Warning	MEDIUM	ACKNOWLEDGED	p. 8
M-02	Price Impact Arbitrage Opportunities	Warning	MEDIUM	ACKNOWLEDGED	p. 9
L-01	Funding Misbehavior	Warning	LOW	ACKNOWLEDGED	p. 10
L-02	Skip Smaller Side May Skip Larger Reserved Side	Warning	LOW	ACKNOWLEDGED	p. 12
L-03	WillPositionCollateralBeSufficient Check Still Relies On Cost Basis	Warning	LOW	ACKNOWLEDGED	p. 13

MEDIUM **M-01**

ACKNOWLEDGED

Funding Charged Is Not Accurate To OI In Tokens

CATEGORY Warning
LOCATION MarketUtils.sol
REVIEW Main Review

— DESCRIPTION

In the `getNextFundingAmountPerSize` flow the `fundingUsd` is calculated based on the `cache.sizeOfPayingSide` which is hardcoded to be the cost basis OI version instead of the open interest of positions in tokens multiplied by the current index token price.

This means that while the `fundingFactorPerSecond` and the `longsPayShorts` value is computed based on the new open interest in tokens method, the actual resulting magnitude of funding that is charged, and the way in which it is collected and distributed from positions, is based on the old cost basis open interest method.

This may be unexpected and can result in scenarios where positions that are very large by way of the size in tokens open interest method actually end up paying a very small amount of funding because of a much smaller cost basis open interest.

— RECOMMENDATION

Currently this deficiency is known, be aware of it.

— RESOLUTION

Status: Acknowledged.

MEDIUM **M-02**

ACKNOWLEDGED

Price Impact Arbitrage Opportunities

CATEGORY Warning
LOCATION Global
REVIEW Main Review

— DESCRIPTION

The update of the open interest measurement type will have an immediate effect on the price impact calculation in all markets.

Some markets will even switch from showing an imbalance towards one direction towards showing it towards the other direction when the measurement switches from a cost basis oriented open interest to a sizeInTokens oriented model.

This may create arbitrage opportunities where traders may foresee that the price impact calculation method is going to change and benefit from opening a long/short when one side is favored as being under-exposed by the cost basis measurement and then closing it when their new position side is now favored as being over-exposed by the new sizeInTokens measurement.

The price impact measurement doesn't even necessarily have to switch sides from one side being seen as the larger side to the other for a sandwiching of opening/closing to be net profitable.

— RECOMMENDATION

Be aware of this possible arbitrage and monitor for it. Ensure that there are no markets which would exhibit a large change in price impact measured imbalance from the open interest measurement update to limit the magnitude of arbitrage possible.

— RESOLUTION

Status: Acknowledged.

LOW **L-01**

ACKNOWLEDGED

Funding Misbehavior

CATEGORY	Warning
LOCATION	MarketUtils.sol
REVIEW	Main Review

DESCRIPTION

In the `getNextFundingFactorPerSecond` function due to the way that the dynamic funding rates are computed, with the `FundingRateChangeType.Decrease` using the exact `fundingDecreaseFactorPerSecond` value and the `FundingRateChangeType.Increase` using the `fundingIncreaseFactorPerSecond` modulated by the `diffUsdToOpenInterestFactor`, it is often possible depending on the active configurations and market conditions that funding would decrease slower if the balance switched from one side to another.

For example, consider the following scenario:

- Longs have OI 100 tokens
- Shorts have OI 50 tokens
- Token price is \$1
- Total OI using the OI in tokens is \$150
- The `fundingExponentFactor` is 1
- $\text{diffUsdToOpenInterestFactor} = \$50 / \$150 = 1/3$
- `fundingDecreaseFactorPerSecond` is 5 (arbitrary)
- `fundingIncreaseFactorPerSecond` is 5 (arbitrary)
- `thresholdForStableFunding` = 2/3
- `thresholdForDecreaseFunding` = 1/6
- Checkpoint A

If a 40 token short is opened from checkpoint A:

- Longs have OI 100 tokens
- Shorts have OI 90 tokens
- Total OI using the OI in tokens is \$190
- $\text{diffUsdToOpenInterestFactor} = 10 / 190 = 1/19$
- We are now below the `thresholdForDecreaseFunding`
- The `nextSavedFundingFactorPerSecond` decreases at a rate of 5 (`fundingDecreaseFactorPerSecond`) per second

If a 60 token short is opened from checkpoint A:

L-01 Funding Misbehavior

CONTINUED

- Longs have OI 100 tokens
- Shorts have OI 110 tokens
- Total OI using the OI in tokens is \$210
- $\text{diffUsdToOpenInterestFactor} = 10 / 210 = 1/21$
- We have now flipped sides and we are using the `FundingRateChangeType.Increase` in the other direction
- Funding now moves in the same direction that the `FundingRateChangeType.Decrease` does, but at a slower rate
- The rate is $5 * 1/21$ per second

As a result, when a more extreme funding change would typically be desired in the market, the funding rate actually moves more slowly to the new paying side and the new paying side actually continues to receive funding for a longer period, paradoxically.

— RECOMMENDATION

Be aware of this behavior when assigning the configurations for the dynamic funding rate.

— RESOLUTION

Status: Acknowledged.

LOW **L-02**

ACKNOWLEDGED

Skip Smaller Side May Skip Larger Reserved Side

CATEGORY Warning
LOCATION MarketUtils.sol
REVIEW Main Review

DESCRIPTION

In the `getBorrowingFactorPerSecond` function, when the `SKIP_BORROWING_FEE_FOR_SMALLER_SIDE` feature is enabled the smaller side as measured by the new `USE_OPEN_INTEREST_IN_TOKENS_FOR_BALANCE` receives zero borrowing fees.

However this may not be the smaller side as measured by reserved USD, since for shorts the reserved USD as calculated by the `getReservedUsd` function is based on the `getOpenInterest` result which gives an open interest in cost basis terms.

RECOMMENDATION

This is not un-similar to the way the skip smaller side feature functioned before when it was based on the cost basis. Just that then it was vice-versa in line with shorts and had a discrepancy with the reservedUsd calculation for longs.

RESOLUTION

Status: Acknowledged.

LOW **L-03**

ACKNOWLEDGED

WillPositionCollateralBeSufficient Check Still Relies On Cost Basis

CATEGORY Warning
LOCATION Global
REVIEW Main Review

DESCRIPTION

The `WillPositionCollateralBeSufficient` validation still relies on the position `sizeInUsd` which is the cost basis form of open interest instead of the current USD value of the `sizeInTokens` of the position to check if the position collateral is sufficient for it's size.

RECOMMENDATION

Consider if this is acceptable for this validation or if it should be updated to be based on the position size in tokens using the current index token price.

RESOLUTION

Status: Acknowledged.

§ 05

Disclaimer.

This report is not, nor should be considered, an "endorsement" or "disapproval" of any particular project or team. This report is not, nor should be considered, an indication of the economics or value of any "product" or "asset" created by any team or project that contracts Guardian to perform a security assessment. This report does not provide any warranty or guarantee regarding the absolute bug-free nature of the technology analyzed, nor do they provide any indication of the technologies proprietors, business, business model or legal compliance.

This report should not be used in any way to make decisions around investment or involvement with any particular project. This report in no way provides investment advice, nor should be leveraged as investment advice of any sort. This report represents an extensive assessing process intending to help our customers increase the quality of their code while reducing the high level of risk presented by cryptographic tokens and blockchain technology.

Blockchain technology and cryptographic assets present a high level of ongoing risk. Guardian's position is that each company and individual are responsible for their own due diligence and continuous security. Guardian's goal is to help reduce the attack vectors and the high level of variance associated with utilizing new and consistently changing technologies, and in no way claims any guarantee of security or functionality of the technology we agree to analyze.

The assessment services provided by Guardian is subject to dependencies and under continuing development. You agree that your access and/or use, including but not limited to any services, reports, and materials, will be at your sole risk on an as-is, where-is, and as-available basis.

Cryptographic tokens are emergent technologies and carry with them high levels of technical risk and uncertainty. The assessment reports could include false positives, false negatives, and other unpredictable results. The services may access, and depend upon, multiple layers of third-parties.

Notice that smart contracts deployed on the blockchain are not resistant from internal/external exploit. Notice that active smart contract owner privileges constitute an elevated impact to any smart contract's safety and security. Therefore, Guardian does not guarantee the explicit security of the audited smart contract, regardless of the verdict.

§ 06

About Guardian.

Guardian is a smart contract security firm. Every engagement is worked by two independent teams competing on the same scope, each with its own tooling and methods, so no single reviewer's blind spot survives.

\$45B+DIGITAL ASSETS UNDER
REVIEW**200+**

ENGAGEMENTS COMPLETED

500+CRITICAL & HIGH FINDINGS
REPORTED

Reach us at audits@guardianaudits.com.